

Udvegadarshini Web App

(उद्वेगदर्शिनी)

Software Application Technical Disclosure & Copyright
Registration Report

INDIAN COPYRIGHT
FILING
SOFTWARE
APPLICATION

1. Software Application Metadata & Authorship

Software Name	Udvegadarshini Web Application (उद्वेगदर्शिनी)
Lead Applicant & Primary Creator	Rajwanth Balaka
Co-Creator	Lakkoju Manobhiram
Academic Institution	Gayatri Vidya Parishad College of Engineering (Autonomous), Visakhapatnam, Andhra Pradesh
Department	Electronics & Communication Engineering (ECE)
Software Architecture	Single-Page Progressive Web Application (PWA / SPA)
Technology Stack	HTML5, CSS3 Glassmorphism, JavaScript ES6+ Engine, Web Serial API, Web Bluetooth (BLE) API, Web Audio API, Chart.js
Live Web App URL	https://rajwanthbalakaofficial-rgb.github.io/Udvegadarshini-App/

2. Executive Summary of the Web Application

Udvegadarshini (उद्वेगदर्शिनी) is an original, multi-functional, web-based healthcare software platform engineered for real-time electroencephalogram (EEG) telemetry visualization, automated stress classification, closed-loop biofeedback relaxation therapy, and clinical health management.

The software operates **driverlessly inside standard web browsers** (Google Chrome, Microsoft Edge) by utilizing modern browser APIs (Web Serial & Web Bluetooth). It processes continuous 250 Hz bio-signal streams, calculates 5 EEG frequency band powers (Delta, Theta, Alpha, Beta, Gamma), computes Hjorth parameters and β/α cognitive stress ratios, and renders dynamic real-time medical dashboards.

3. Complete Features & Software Modules Inventory

- **Real-Time Telemetry & Oscilloscope Engine (`app.js`)**: Driverless Web Serial/BLE integration, SVG Dynamic Gauge (0–100% Score, 5 Clinical States), Delta-Gamma Band Power Bars, and Chart.js rolling oscilloscope graphs.
- **Web Audio Sound Therapy Synthesizer (`audio-frequency.js`)**: Native Web Audio API synthesizer for 432Hz Miracle Tone, 528Hz Transformation Tone, 396Hz Fear Release Tone, 10Hz Alpha Binaural Beats, 6Hz Theta Beats, and Rain Soundscape.
- **Guided 4–7–8 Biofeedback Breathing Visualizer**: Interactive animated breathing circle (4s Inhale, 7s Hold, 8s Exhale) for vagus nerve stimulation.
- **5 Stress-Relief Mini-Games (`games.js`)**: Bubble Wrap Popper, Arrow Logic Flow, Sudoku Mind Focus, Memory Pattern Match, and 2048 Mind Merge.
- **Āpta AI (आप्त AI) Conversational Assistant (`apta-ai.js`)**: Multi-lingual empathetic AI supporting 100% Tanglish (Teluglish), English, Telugu, and Hindi with interactive 1–tap action buttons in speech bubbles.
- **Clinical Management Portal & Medical PDF Export**: Multi-tenant Role-Based Access Control (Patient, Doctor, Hospital Admin, Master Admin) with automated PDF medical session report generator.

4. Comparative Software Analysis against Commercial Web & Mobile Apps

Feature / Metric	Mindfulness Apps (Calm / Headspace)	Telemetry Apps (Welltory / HeartMath)	Headset Apps (Muse / NeuroSky)	Udvegadarshini Web App (Proposed)
App Access	Mobile App download	Mobile App only	Desktop/Mobile Native App	Progressive Web App (Zero-Install Browser Access)
Driver Requirement	None (No real-time EEG)	Proprietary Bluetooth drivers	Native OS drivers required	Zero-Driver Web Serial & Web BLE

Feature / Metric	Mindfulness Apps (Calm / Headspace)	Telemetry Apps (Welltory / HeartMath)	Headset Apps (Muse / NeuroSky)	Udvegadarshini Web App (Proposed)
Live EEG Telemetry	No EEG (Audio only)	No EEG (Optical HRV only)	Live EEG display	Live 5-Band EEG Power Bars + Waveform Oscilloscope
Sound Generation	Pre-recorded MP3 files	None	Basic audio tracks	Native Web Audio API Synthesizer (432Hz/528Hz)
Mini-Games	None	None	None	5 Built-in Tactile & Logic Mini-Games
AI Companion	Basic static chatbot	Text insights	None	Āpta AI: Multi-lingual & 100% Tanglish Engine
PDF Export	None	Limited history	Raw CSV only	Full Hospital Roster + Instant PDF Session Export

5. Copyright Protection Claims (Software Source Code & GUI)

The applicants claim exclusive copyright ownership under the Indian Copyright Act, 1957 for the following original software assets:

1. `app.js` : Master JavaScript Engine for Web Serial/BLE telemetry, SVG gauge rendering, Chart.js graphs, and RBAC portal logic.
2. `apta-ai.js` : Āpta AI Conversational Engine, intent recognition algorithms, Tanglish dialogue rules, and chat action buttons.
3. `audio-frequency.js` : Web Audio API Solfeggio frequency oscillator synthesizer and binaural beats signal generator.
4. `games.js` : 5 HTML5 Canvas & DOM Mini-Games Implementation.
5. `i18n.js` : Multi-language translation engine and dynamic dictionary.
6. `index.html` & `styles.css` : Original Graphical User Interface (GUI) layout, glassmorphism styling, and responsive single-page design.

Rajwanth Balaka

Lead Applicant & Primary Software Architect

Lakkoju Manobhiram

Co-Creator

Udveadarshini Software Technical Disclosure | Department of ECE, Gayatri Vidya Parishad College of Engineering (Autonomous)